

3.2 Hydrocarbons

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.2 Hydrocarbons
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The cracking of a single hydrocarbon molecule, C_nH_{2n+2} , produces two hydrocarbon molecules only. Each hydrocarbon product contains the same number of carbon atoms in one molecule. Each hydrocarbon product has non-cyclic structural isomers.

What is the value of n ?

- A. 9
- B. 8
- C. 6
- D. 4

[1 mark]

Question 2

Which reaction would **not** give propene?

- A. Adding excess hot concentrated sulfuric acid to propan-1-ol.
- B. Adding warm aqueous sodium hydroxide to 2-bromopropane.
- C. Adding warm ethanolic sodium hydroxide to 1-bromopropane.
- D. Passing propan-2-ol vapour over heated aluminium oxide.

[1 mark]

Question 3

Ethene reacts with aqueous bromine to give two products, CH_2BrCH_2Br and CH_2BrCH_2OH .

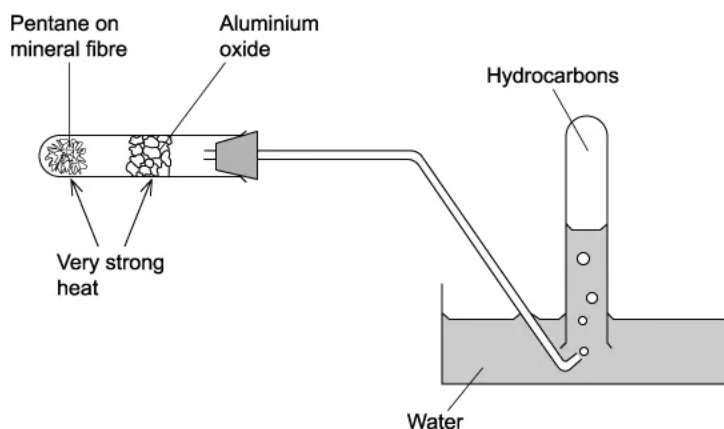
Which statement about these products is correct?

- A. Both products are obtained in this reaction by electrophilic substitution.
- B. Both products are obtained in this reaction by nucleophilic addition.
- C. Both products can be hydrolysed to form the same organic compound.
- D. Both products can form hydrogen bonds with water.

[1 mark]

Question 4

Which hydrocarbon would **not** be collected in the inverted tube by heating pentane, $\text{CH}_3(\text{CH}_2)_3\text{CH}_3$, in the apparatus shown?



- A. CH_4
- B. CH_3CH_3
- C. $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$
- D. $\text{CH}_3(\text{CH}_2)_8\text{CH}_3$

[1 mark]

Question 5

Pent-2-ene, $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$, reacts in a similar way to ethene.

Pent-2-ene is reacted with cold, dilute, acidified manganate (VII) ions.

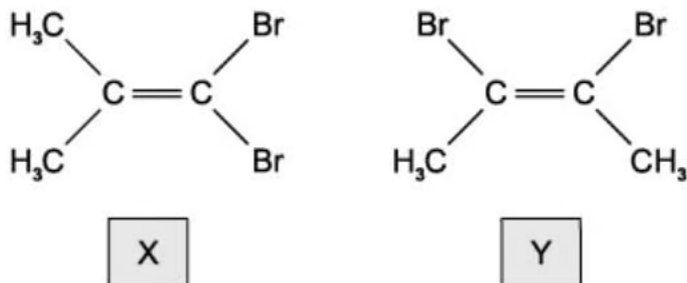
What will be produced in the greatest amount?

- A. $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$
- B. $\text{CH}_3\text{CH}_2\text{COCOCH}_3$
- C. a mixture of $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- D. $\text{CH}_3\text{CH}_2\text{COOH}$ and CH_3COOH

[1 mark]

Question 6

Isomers **X** and **Y** both react with HBr.



A mixture of **X** and **Y** is reacted with HBr.

Which three structures represent three **different** possible products of this reaction?

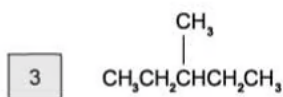
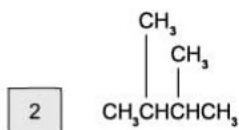
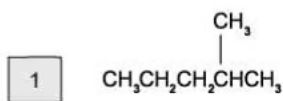
A	$(\text{CH}_3)_2\text{CHCBr}_3$	$(\text{CH}_3)_2\text{CBrCHBr}_2$	$\text{CH}_3\text{CHBrCHBrCH}_3$
B	$(\text{CH}_3)_2\text{CHCBr}_3$	$(\text{CH}_3)_2\text{CBrCHBr}_2$	$\text{CH}_3\text{CBr}_2\text{CHBrCH}_3$
C	$(\text{CH}_3)_2\text{CBrCBr}_3$	$(\text{CH}_3)_2\text{CHCBr}_3$	$\text{CH}_3\text{CBr}_2\text{CHBrCH}_3$
D	$(\text{CH}_3)_2\text{CBrCHBr}_2$	$\text{CHBr}_2\text{CBr}(\text{CH}_3)\text{CH}_3$	$\text{CH}_3\text{CHBrCBr}_2\text{CH}_3$

[1 mark]

Question 7

During the bromination of methane, the free radical $\text{CH}_3\bullet$ is generated and a possible terminating step of this reaction is the formation of C_2H_6 by the combination of two free radicals.

What could be produced in a terminating step during the bromination of propane?



- A. 1 and 2
- B. 2 and 3
- C. 1 only
- D. 2 only

[1 mark]

Question 8

Which alkenes, on reaction with steam at 600 K and 6×10^6 Pa pressure in the presence of a phosphoric acid catalyst, could produce an alcohol containing a chiral carbon atom?

- 1 $(\text{CH}_3)_2\text{C}=\text{CH}_2$
- 2 $\text{CH}_3\text{CH}=\text{CHCH}_3$
- 3 $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$

- A. 1, 2 and 3
- B. 1 and 2
- C. 2 and 3
- D. 1 only

[1 mark]**Question 9**

0.200 mol of a hydrocarbon undergoes complete combustion to give 35.2 g of carbon dioxide and 14.4 g of water as the only products.

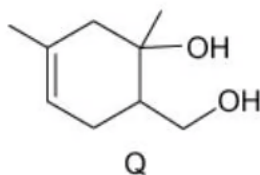
What is the molecular formula of the hydrocarbon?

- A. C_2H_4
- B. C_2H_6
- C. C_4H_4
- D. C_4H_8

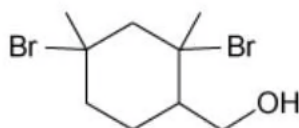
[1 mark]

Question 10

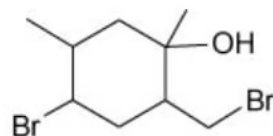
What is the major product formed when compound Q is warmed with excess HBr?



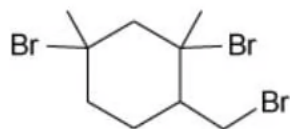
A.



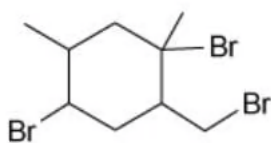
B.



C.



D.



[1 mark]

Model Answers: Medium

1

The correct answer is **B** because:

- Cracking is a reaction in which the larger saturated hydrocarbon molecules (alkanes) in crude oil are broken down into smaller, more useful hydrocarbon molecules, including both alkanes and alkenes.
- Consider the alkane with $n = 8$, C_8H_{18}
- We are told each hydrocarbon product contains the same number of carbon atoms in one molecule, so, the cracked molecules will have 4 carbon atoms (butene).
- We are also told each hydrocarbon product has non-cyclic structural isomers, which butene does:
 - But-1-ene
 - cis-but-2-ene
 - trans-but-2-ene
 - 2-methylprop-1-ene (isobutene)
- Therefore, we can conclude that $n = 8$.

A is incorrect as we are told each molecule produced contains the same number of carbon atoms, so n must be even.

C is incorrect as when $n = 6$, the cracked molecules will have 3 carbon atoms (propene), this alkene has two isomers, one of which is cyclopropane, which is a cyclic isomer.

D is incorrect as when $n = 4$, the cracked molecules will be ethene, which has no isomers.

2

The correct answer is **B** because:

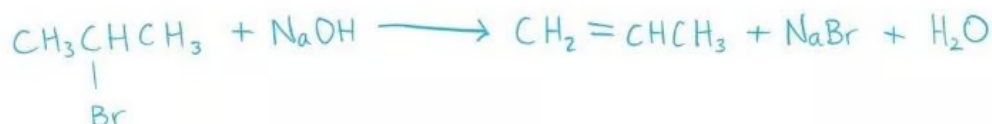
- Adding warm aqueous sodium hydroxide to 2-bromopropane will give propan-2-ol.
- This is a nucleophilic substitution reaction.



A is incorrect as adding excess hot concentrated sulfuric acid to propan-1-ol gives propene (dehydration reaction):



C is incorrect as adding warm ethanolic sodium hydroxide to 1-bromopropane gives propene (elimination reaction):



D is incorrect as passing propan-2-ol vapour over heated aluminium oxide gives propene (dehydration reaction):

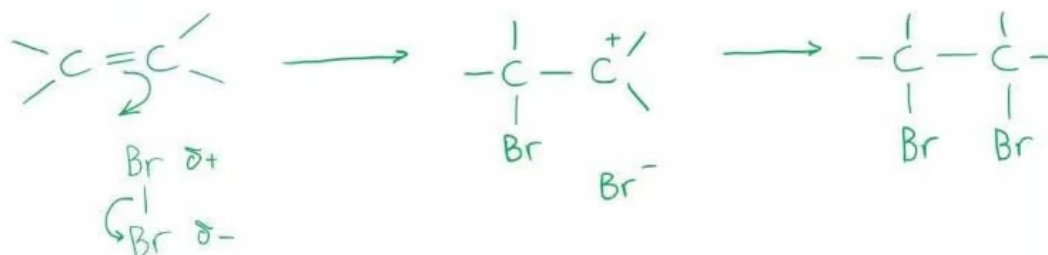


3

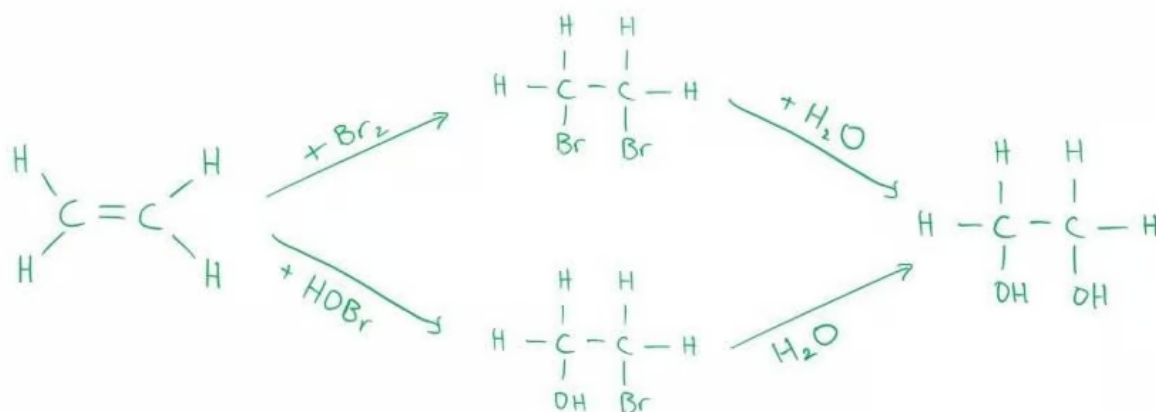
The correct answer is **C** because:

- Bromine is an electrophile (electron loving species).
- It reacts with ethene via electrophilic addition.

Electrophilic addition mechanism:



- The reaction between ethene and bromine forms 1,2-dibromoethane ($\text{CH}_2\text{BrCH}_2\text{Br}$) and 2-bromoethanol ($\text{CH}_2\text{BrCH}_2\text{OH}$).



- When combined with water, both products will react to make ethane-1,2-diol.
 - Therefore, **C** is correct.

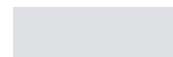
A & B are incorrect as both products are obtained in this reaction by electrophilic addition.

D is incorrect as hydrogen bonds form between a hydrogen atom covalently bonded to a very electronegative atom such as a N, O, or F. Therefore, 2-bromoethanol ($\text{CH}_2\text{BrCH}_2\text{OH}$) forms hydrogen bonds with water whereas 1,2-dibromoethane ($\text{CH}_2\text{BrCH}_2\text{Br}$) does not.

4

The correct answer is **D** because:

- Heating pentane in an inverted tube over aluminium oxide is one way to crack the molecule.
- Cracking is a reaction in which the larger saturated hydrocarbon molecules (alkanes) are broken down into smaller, more useful hydrocarbon molecules, including both alkanes and alkenes.
- Typically, in cracking reactions, a mixture of different alkanes and alkenes are produced.



4

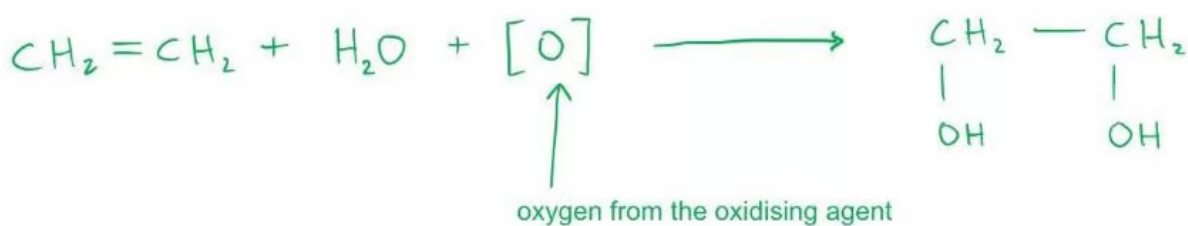
The correct answer is **D** because:

- Heating pentane in an inverted tube over aluminium oxide is one way to crack the molecule.
- Cracking is a reaction in which the larger saturated hydrocarbon molecules (alkanes) are broken down into smaller, more useful hydrocarbon molecules, including both alkanes and alkenes.
- Typically, in cracking reactions, a mixture of different alkanes and alkenes are produced.
- So, for pentane, the cracking process will produce alkanes and alkenes of five or less carbons.
 - Therefore, the compound $\text{CH}_3(\text{CH}_2)_8\text{CH}_3$ would not be produced in this reaction as it contains more carbon atoms than the original pentane.

5

The correct answer is **A** because:

- When ethene reacts with **cold, dilute**, acidified manganate (VII) ions, ethane-1,2-diol is produced.



- So, in a similar way, when pent-2-ene reacts with cold, dilute, acidified manganate (VII) ions, pentane-1,2-diol is produced:
 - $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3 + \text{H}_2\text{O} + [\text{O}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$

B is incorrect as $\text{CH}_3\text{CH}_2\text{COCOCH}_3$ is 2,3-pentanedione this cannot be synthesised from pent-2-ene.

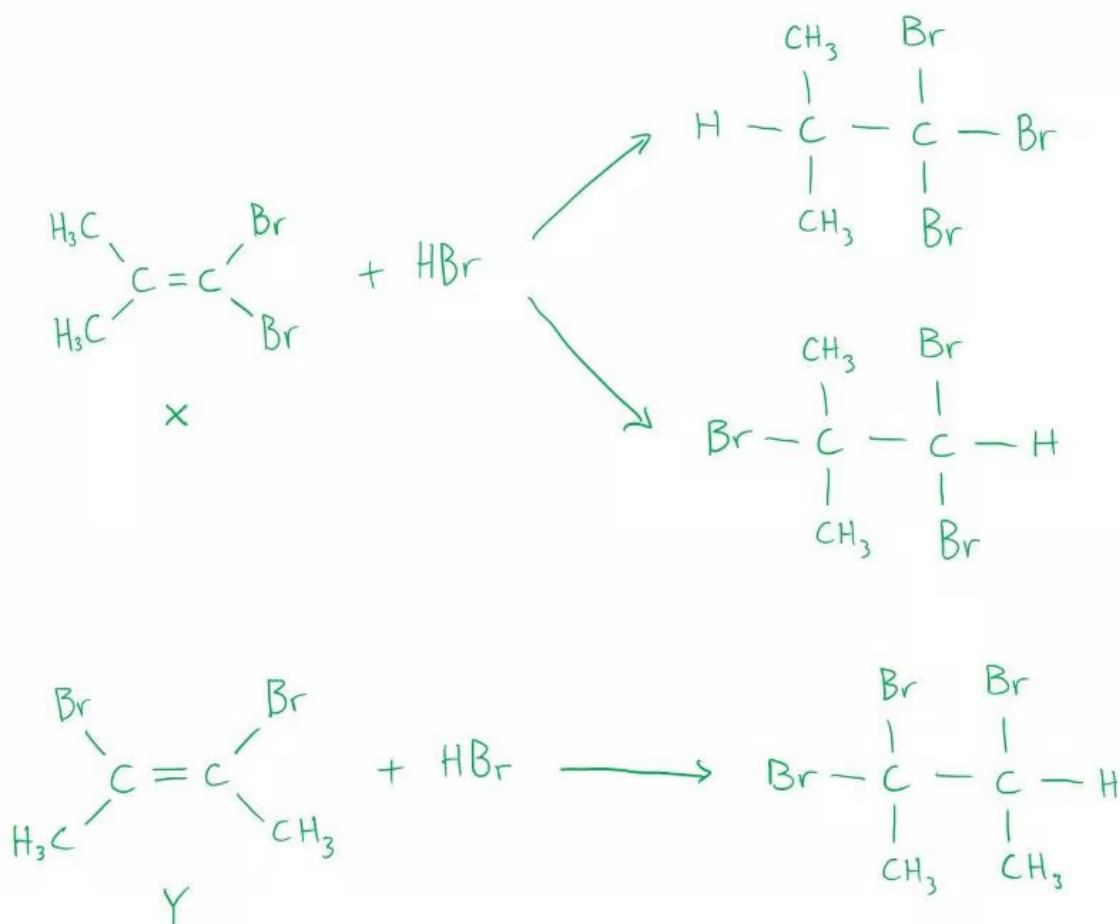
C is incorrect as $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ is pentan-3-ol and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$ is pentan-2-ol. A mixture of pentan-2-ol and pentan-3-ol can be synthesised from **hydration** of pent-2-ene.

D is incorrect as $\text{CH}_3\text{CH}_2\text{COOH}$ is propanoic acid and CH_3COOH is ethanoic acid. These could only be produced with **hot, concentrated**, manganate (VII) ions.

6

The correct answer is **B** because:

- Bromine is an electrophile (electron loving species).
- It reacts with the alkene isomers via electrophilic addition by breaking the $\text{C}=\text{C}$ double bond.
- The best way to approach a question like this is to completely draw out the structural formula of each product:



- So, the reaction between HBr and isomers **X** and **Y** form:
 - $(\text{CH}_3)_2\text{CHCBr}_3$
 - $(\text{CH}_3)_2\text{CBrCHBr}_2$
 - $\text{CH}_3\text{CBr}_2\text{CHBrCH}_3$

7

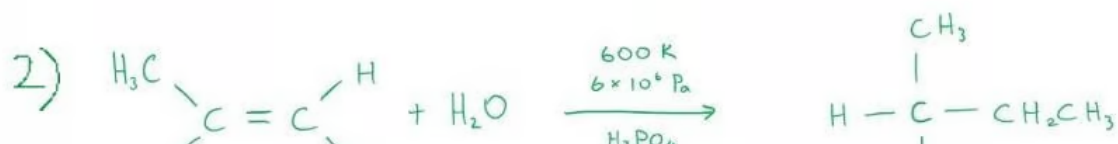
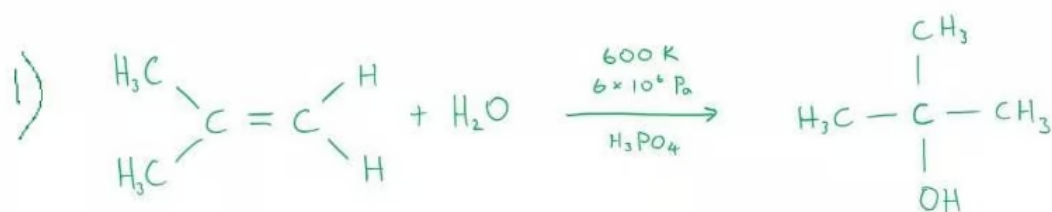
The correct answer is **D** because:

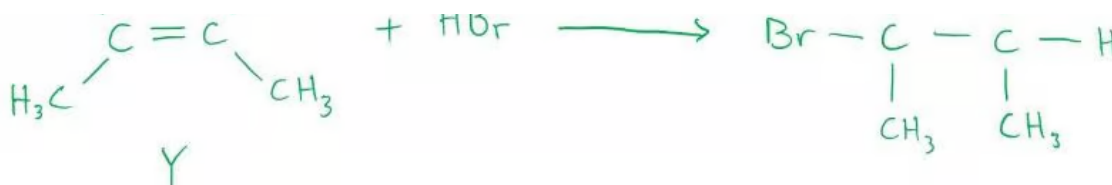
- Initiation step:
 - $\text{Br}_2 \rightarrow 2\text{Br}\cdot$
- Possible propagation steps:
 - $\text{CH}_3\text{CH}_2\text{CH}_3 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\cdot + \text{HBr}$
 - $\text{CH}_3\text{CH}_2\text{CH}_3 \rightarrow (\text{CH}_3)_2\text{CH}\cdot + \text{HBr}$
- So, there are two free radicals generated in bromination of propane.
 - $(\text{CH}_3)_2\text{CH}\cdot$ and $\text{CH}_3\text{CH}_2\text{CH}_2\cdot$
- However, $(\text{CH}_3)_2\text{CH}\cdot$ is more stable than $\text{CH}_3\text{CH}_2\text{CH}_2\cdot$ so the reaction proceeds via $(\text{CH}_3)_2\text{CH}\cdot$ radical.
- So, at termination the product formed will be:
 - $(\text{CH}_3)_2\text{CH}\cdot + (\text{CH}_3)_2\text{CH}\cdot \rightarrow (\text{CH}_3)_2\text{CH}-\text{CH}(\text{CH}_3)_2$

8

The correct answer is **C** because:

- A chiral carbon atom is a carbon atom that is attached to four different types of atoms or groups of atoms.
- When each alkene is mixed with 600 K and 6×10^6 Pa steam and passed over a catalyst consisting of solid silicon dioxide coated with phosphoric (V) acid, the double bond opens and allows the -OH group to join the molecule.
- The best way to approach this question is to draw out each resulting alcohol molecule:





- So, the reaction between HBr and isomers **X** and **Y** form:
 - $(\text{CH}_3)_2\text{CHCBr}_3$
 - $(\text{CH}_3)_2\text{CBrCHBr}_2$
 - $\text{CH}_3\text{CBr}_2\text{CHBrCH}_3$

7

The correct answer is **D** because:

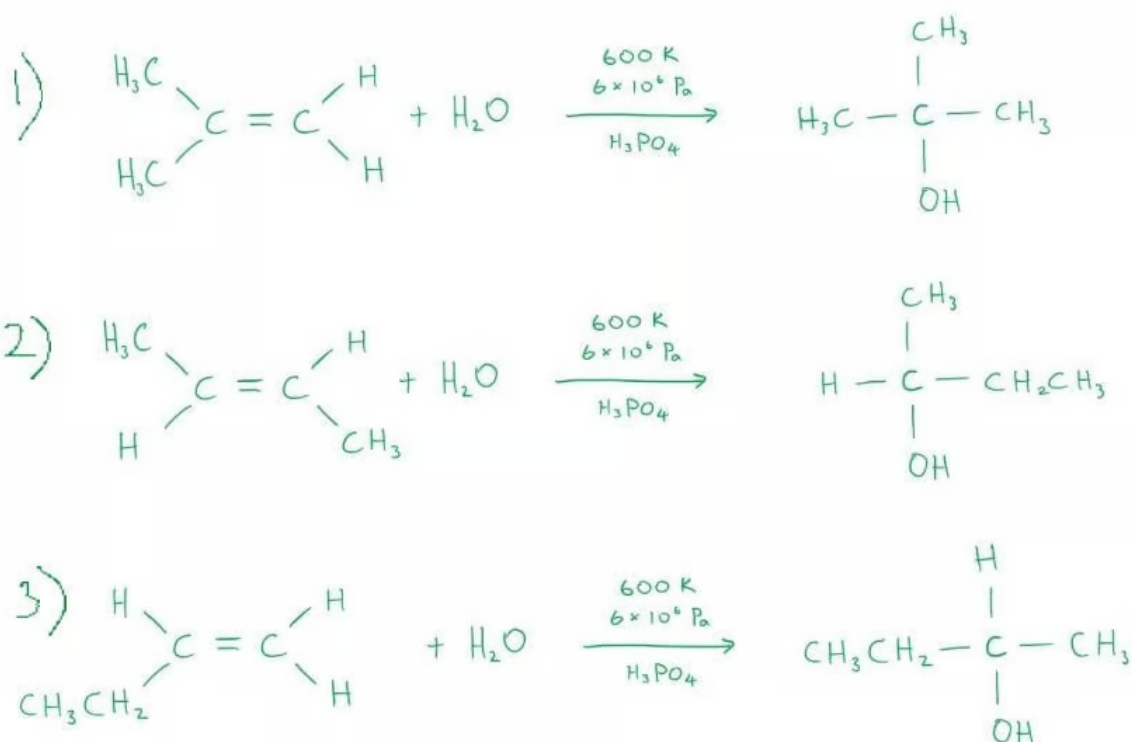
- Initiation step:
 - $\text{Br}_2 \rightarrow 2\text{Br}\cdot$
- Possible propagation steps:
 - $\text{CH}_3\text{CH}_2\text{CH}_3 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\cdot + \text{HBr}$
 - $\text{CH}_3\text{CH}_2\text{CH}_3 \rightarrow (\text{CH}_3)_2\text{CH}\cdot + \text{HBr}$
- So, there are two free radicals generated in bromination of propane.
 - $(\text{CH}_3)_2\text{CH}\cdot$ and $\text{CH}_3\text{CH}_2\text{CH}_2\cdot$
- However, $(\text{CH}_3)_2\text{CH}\cdot$ is more stable than $\text{CH}_3\text{CH}_2\text{CH}_2\cdot$ so the reaction proceeds via $(\text{CH}_3)_2\text{CH}\cdot$ radical.
- So, at termination the product formed will be:
 - $(\text{CH}_3)_2\text{CH}\cdot + (\text{CH}_3)_2\text{CH}\cdot \rightarrow (\text{CH}_3)_2\text{CH}-\text{CH}(\text{CH}_3)_2$

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The correct answer is **C** because:

- A chiral carbon atom is a carbon atom that is attached to four different types of atoms or groups of atoms.
- When each alkene is mixed with 600 K and 6×10^6 Pa steam and passed over a catalyst consisting of solid silicon dioxide coated with phosphoric (V) acid, the double bond opens and allows the -OH group to join the molecule.

- The best way to approach this question is to draw out each resulting alcohol molecule:



- Reactions 2 and 3 produce alcohols containing a chiral carbon atom, but not reaction 1.
 - So, option **C** is correct.

9

The correct answer is **D** because:

- The equation for combustion is:
 - $\text{C}_x\text{H}_y + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
- Number of moles of CO_2 :

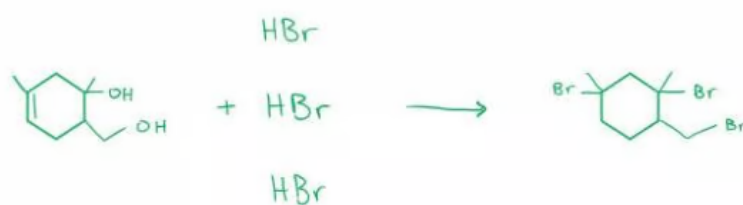
$$\frac{m}{M_r} = \frac{35.2 \text{ g}}{44}$$

- $\text{Moles} = \frac{m}{M_r} = \frac{14.4 \text{ g}}{(12+16+16) \text{ g mol}^{-1}} = 0.8 \text{ mol}$
- H_2O has the same number of moles:
 - $\text{Moles} = \frac{m}{M_r} = \frac{14.4 \text{ g}}{(1+1+16) \text{ g mol}^{-1}} = 0.8 \text{ mol}$
- C_xH_y has 0.2 mol, which is $\frac{1}{4}$ of the moles of CO_2 and H_2O , so:
 - $\text{C}_x\text{H}_y + \text{O}_2 \rightarrow 4\text{CO}_2 + 4\text{H}_2\text{O}$
- Balancing the Cs, and following the molar ratio, x is 4, and y is 8.
- The O_2 reactant must be balanced so it has 6, therefore:
 - $\text{C}_4\text{H}_8 + 6\text{O}_2 \rightarrow 4\text{CO}_2 + 4\text{H}_2\text{O}$
- Therefore, the answer is C_4H_8 (butane).

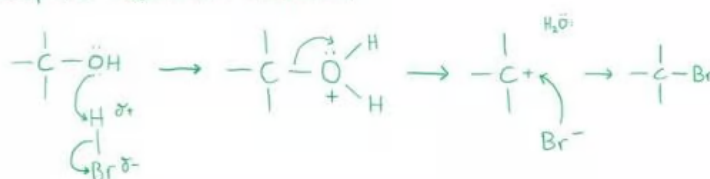
10

The correct answer is **C** because:

- The $-\text{OH}$ group will react with HBr to undergo nucleophilic substitution.
- Each $-\text{OH}$ will be replaced with Br .
- The double bond in the compound will also react with HBr and undergo electrophilic addition.
 - Hence **C** will be the major product formed.
 -



nucleophilic substitution mechanism



Electrophilic addition mechanism:

